

Final Report: PermitPulse

Introduction

Toronto's rapid population growth and persistent housing affordability crisis have intensified the need to understand the factors driving construction costs. With the average home price exceeding \$1.1 million in 2025 (CP24, 2025), policymakers, developers, and investors face pressure to deliver projects efficiently while managing costs. Construction expenses are shaped by regulatory, structural, and geographic factors, making it essential to identify their relative influence for informed urban planning.

This study addresses the question: *What key factors influence the estimated construction cost of building permits in Toronto between 2017 and 2024?* Using over 100,000 cleared building permit records from the City of Toronto's open data portal, we examine predictors including permit type, structure type, ward location, number of dwelling units created, and residential floor area.

Prior research highlights the relevance of these factors. Elhag et al. (2005) found that location and gross floor area significantly predict construction costs, with log-linear models improving fit. Kim et al. (2012) showed that project type and building specifications influence cost estimation accuracy, while Liu and Zhu (2021) emphasized how regional differences and scale affect cost variability in metropolitan areas.

Multiple linear regression is appropriate here as it models the relationship between a continuous outcome, estimated construction cost, and multiple categorical and continuous predictors while controlling for confounders. This approach enables hypothesis testing, interpretable coefficient estimates, and thorough assumption diagnostics.

By integrating municipal data with modelling methods, this research aims to produce insights for improving cost efficiency and guiding policy in Toronto's evolving housing market.

Methods

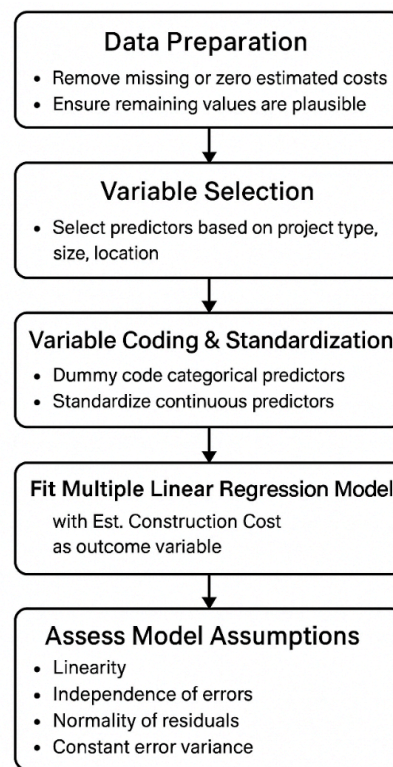


Figure 1. Simplified analysis workflow for predicting construction costs from Toronto building permit data .

Our goal was to develop a statistical model capable of predicting the estimated construction cost of building permits in Toronto using information available in the permits themselves. The analysis was based on the City of Toronto’s Building Permits – Cleared Permits dataset from 2017 to 2024. We removed records with missing essential information or an estimated cost of zero, as these would not contribute meaningful information. We also ensured that any remaining values were consistent with plausible real-world ranges to minimize the impact of data entry errors. The outcome variable was Estimated Construction Cost, and the predictors included: PERMIT_TYPE (categorical), STRUCTURE_TYPE (categorical), WARD_GRID (categorical location), DWELLING_UNITS_CREATED (numeric), and RESIDENTIAL (numeric floor area). These predictors were chosen because they reflect a project’s type, size, and location.

Categorical predictors, such as PERMIT_TYPE and STRUCTURE_TYPE, were converted into binary indicator variables (“dummy variables”), with one category set as the reference group. Continuous variables, including RESIDENTIAL, were standardized to have a mean of zero and a standard deviation of one. This standardization improves model stability and ensures that predictors measured on different scales can be compared fairly.

We used multiple linear regression to examine the relationship between estimated construction cost and the selected predictors. This method allows estimation of the effect of each predictor while holding others constant and supports hypothesis testing for statistical significance.

In line with best practices discussed in lectures, we assessed the core assumptions of multiple linear regression:

1. Linearity: The relationship between predictors and the response variable should be approximately linear. This was assessed visually using residual plots.
2. Independence of errors: Residuals should not be correlated with one another, which was assumed given the independence of individual building permits.
3. Normality of residuals: Residuals should follow an approximately normal distribution, assessed using Q–Q plots.
4. Constant variance (homoscedasticity): The spread of residuals should remain consistent across fitted values, evaluated through Scale–Location plots.

If diagnostic plots suggested deviations from these assumptions, we applied the Box–Cox procedure (via the *MASS* package in R) to determine the optimal transformation of the response variable. The Box–Cox method evaluates a range of power transformations to identify one that best stabilizes variance and improves the normality of residuals. When the suggested transformation parameter was close to zero, we used a natural logarithm transformation, as it is both statistically effective and interpretable for proportional cost changes. This transformation choice aligns with previous research on construction cost modelling, where log-linear models often improve both interpretability and statistical performance (Elhag et al., 2005).

The final model was evaluated using the adjusted R^2 statistic to measure explanatory power while accounting for the number of predictors. Statistical significance of predictors was assessed using p-values, and coefficients from the log-linear model were exponentiated to express results in multiplicative terms (e.g., cost ratios).

All data cleaning, transformation, modelling, and visualization were conducted in R. Key packages included *dplyr* for data manipulation, *ggplot2* for creating diagnostic and exploratory visualizations, *MASS* for transformation selection, and *broom* for tidying model outputs. The workflow was documented in an R Markdown file, ensuring all steps were reproducible and well-documented for transparency.

Results

Following the data cleaning and preparation described in the “methods” section, we began by fitting a multiple linear regression model with `EST_CONST_COST` as the response variable and predictors `PERMIT_TYPE`, `STRUCTURE_TYPE`, `WARD_GRID`, `DWELLING_UNITS_CREATED`, and `RESIDENTIAL` (floor area). This initial model used untransformed construction costs.

The preliminary model output indicated that several predictors were statistically significant ($p < 0.01$). However, diagnostic plots revealed substantial violations of regression assumptions—particularly non-normal residuals and heteroscedasticity—indicating the need for transformation. To evaluate possible transformations, we applied the Box–Cox procedure to the initial model (Figure 2).

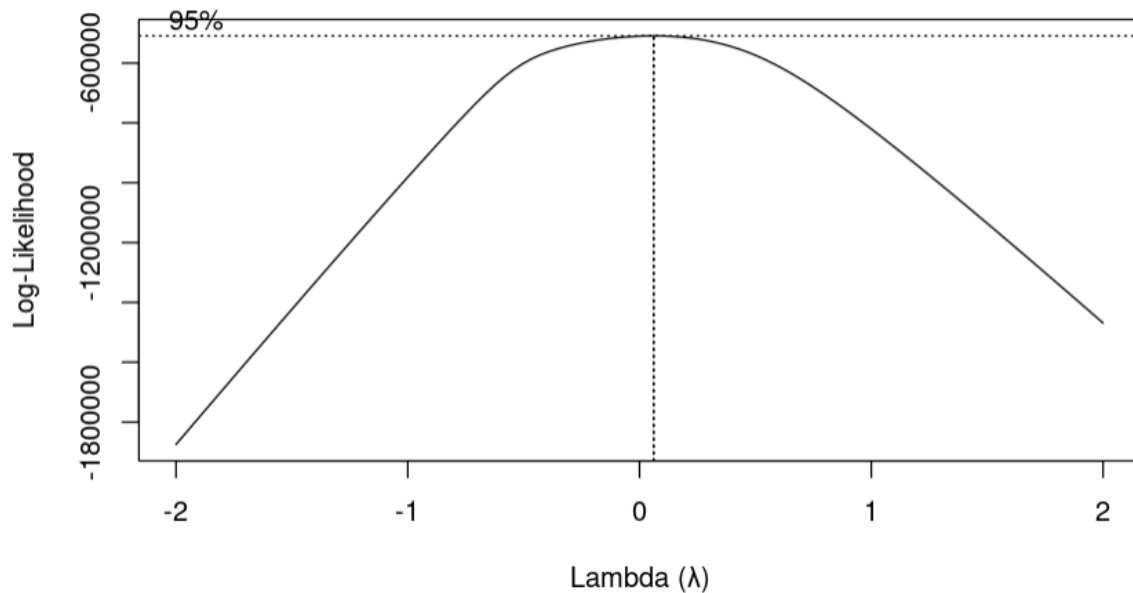


Figure 2: Box-Cox Transformation for Estimated Construction Cost

The optimal λ value of 0.1 (95% CI) was close to zero, indicating that a logarithmic transformation would address the skewness in EST_CONST_COST, where most permits have low costs but a few are extremely high. The choice to log-transform rather than apply other nonlinear transformations, such as square root or inverse, was guided by both statistical evidence and interpretability. A log-linear model allows for coefficients to be interpreted in terms of percentage changes or multiplicative effects. Following the Box–Cox results, we transformed the response variable as $\log(\text{EST_CONST_COST})$. The resulting distribution is shown in **Figure 3**.

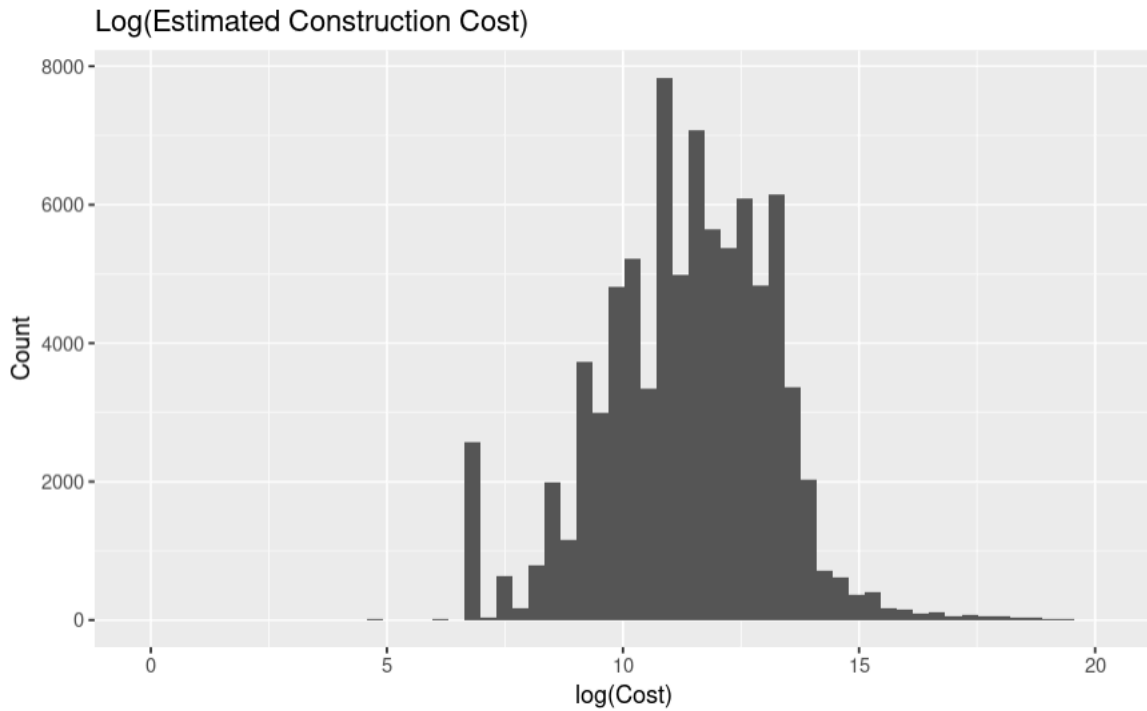


Figure 3: Histogram of Log-Transformed Estimated Construction Costs

The log transformation reduced skewness, compressed extreme values, and lessened the influence of large projects. This change was expected to improve the model's adherence to the linear regression assumptions of constant variance and normally distributed residuals. We examined the log-transformed costs across different permit categories to assess variation by project type. **Figure 4** compares the ten most common permit types in the dataset.

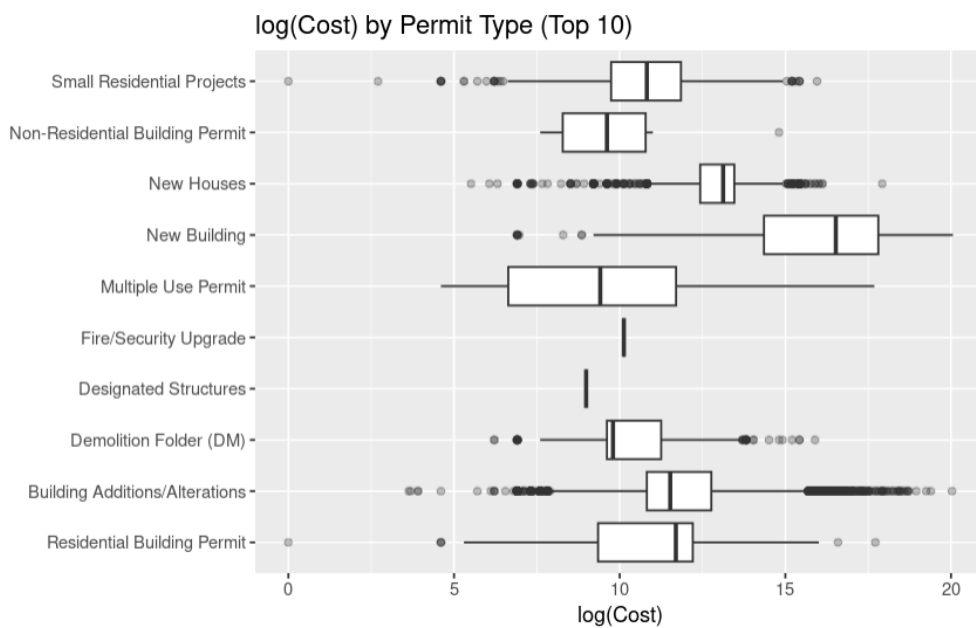


Figure 4: Boxplot of Log-Transformed Costs by Permit (top 10)

The above figure illustrates Boxplot distribution of log-transformed costs based on top 10 permit types. It is evident that there are differences in median costs between various categories. Consider “New Building” and “Multiple Use Permits”; they show higher median log costs when compared to “Small Residential Projects” and “Fire/Security Upgrade”. These patterns confirm PERMIT_TYPE as a key predictor, with clear separation between high- and low-cost categories, justifying its inclusion in the final model.

After transformation, we refit the model using $\log(\text{EST_CONST_COST})$ as the response variable. The final fitted model took the following form:

$$\log(\text{EST_CONST_COST}) = \beta_0 + \beta_1(\text{PERMIT_TYPE}) + \beta_2(\text{STRUCTURE_TYPE}) + \beta_3(\text{WARD_GRID}) + \beta_4(\text{DWELLING_UNITS_CREATED}) + \beta_5(\text{RESIDENTIAL}) + \epsilon$$

where the categorical variables PERMIT_TYPE, STRUCTURE_TYPE, and WARD_GRID are coded relative to their respective baseline categories.

A comparison of diagnostic plots between the initial and log-transformed models (Appendix A, Figures 5 and 6) shows clear improvements in assumption adherence. In the initial model, residuals displayed substantial right-skew, heteroscedasticity, and several extreme high-leverage points. After applying the natural logarithm transformation, residuals were more symmetrically distributed, the spread of residuals became more consistent across fitted values, and the number and influence of outliers were reduced.

The final model explained approximately 53% of the variance in logged costs (Adjusted $R^2 = 0.532$), an improvement over the untransformed model’s performance. The adjusted R^2 value of 0.532 suggests that while a large portion of cost variability is captured by the model, other unmeasured factors—such as contractor-specific pricing or site-specific conditions—may still play a role.

Term	Estimate	Std. Error	t value	p-value	95% CI (Low)	95% CI (High)
(Intercept)	10.716	0.113	94.972	0.000	10.495	10.937
Building Additions/Alterations	0.741	0.094	7.882	0.000	0.556	0.925
Demolition Folder (DM)	-0.586	0.111	-5.281	0.000	-0.803	-0.368
Multiple Use Permit	-2.316	0.542	-4.276	0.000	-3.378	-1.255
New Building	3.606	0.123	29.288	0.000	3.365	3.847
New Houses	2.056	0.090	22.896	0.000	1.880	2.232
Small Residential Projects	-0.318	0.089	-3.566	0.000	-0.493	-0.143
Apartment Building	-0.780	0.052	-14.893	0.000	-0.883	-0.677
Multiple Unit Building	-0.579	0.053	-11.003	0.000	-0.682	-0.476
Restaurant 30 Seats or Less	-0.703	0.059	-11.936	0.000	-0.818	-0.587

Table 1. Coefficients for $\log(EST_CONST_COST)$ model with 95% confidence intervals.

Interpreting coefficients in a log-linear model provides multiplicative effects:

- New Building permits were associated with costs approximately $36.9\times$ higher than the baseline “Residential Building Permit” ($\exp(3.606) \approx 36.9$), holding all else constant.
- New Houses cost about $7.8\times$ more than the baseline.
- Small Residential Projects cost about 27% less ($\exp(-0.318) \approx 0.73$) than the baseline.
- Demolition Folder (DM) permits cost about 44% less than baseline.
- Certain structure types, such as “Apartment Building” and “Multiple Unit Building,” were associated with lower logged costs compared to the baseline structure type.

Diagnostic plots for the log-transformed model (Appendix A, Figures 6) indicated improved residual behaviour. The Residuals vs Fitted plot showed reduced curvature and more even spread, the Normal Q–Q plot showed residuals closer to the diagonal, and the Scale–Location plot confirmed improved homoscedasticity. Only a small number of high-leverage points remained influential.

Transforming the response variable using the natural logarithm improved model fit and better satisfied regression assumptions. The final model explains more than half of the variation in construction costs and provides clear evidence that costs are significantly influenced by:

1. Permit type: New Building permits are far more expensive than most other categories, while small residential projects and demolition permits cost substantially less.
2. Structure type: Certain building forms, such as apartment and multi-unit buildings, are less costly than the baseline structure type.
3. Location: Wards differ significantly in average construction costs, reflecting varying land values, demand, and development regulations.
4. Project scale: Larger floor areas and higher dwelling unit counts strongly predict higher costs.

Conclusion & Limitations

This study examined the key factors influencing the estimated construction cost of building permits issued in Toronto between 2017 and 2024. Using over 100,000 records from the City of Toronto's cleared building permits dataset, we applied multiple linear regression to assess the impact of permit type, structure type, ward location, dwelling units created, and residential floor area on costs.

The final log-linear regression model explained about 53% of cost variation, indicating that the selected predictors capture a substantial share of variability. Permit type emerged as the most influential factor: for example, "New Building" permits were associated with costs about 36.9 times greater than the baseline "Residential Building Permit" ($\exp(3.606) \approx 36.9$), holding other variables constant. Larger project scale, measured by dwelling units created and floor area, also significantly increased costs, while certain structure types, such as apartment buildings, were associated with lower average costs relative to the baseline.

These findings are aligned with the literature that identifies project type, location, and size as fundamental cost drivers (Elhag et al., 2005; Kim et al., 2012; Liu & Zhu, 2021). Significant ward-level differences reflect Toronto's heterogeneous land values, regulatory environment, and infrastructure limitations. From a policy-making perspective, cost-containment measures are to be project-type and location-specific. The results can be used by developers and investors for improving budgeting, risk analysis, and feasibility.

There are limitations worth mentioning. Firstly, the data set lacked variables for contractor experience, materials price volatility, labour market availability, and project site conditions that add to variation in cost. Secondly, although the Box-Cox transformation assisted in improving assumption conformity there were nevertheless a few high-leverage points that might represent unusually large projects that influence model estimates. Thirdly, ward-level location data afford only rough geographic resolution; greater spatial detail may

show more subtle location effects. Finally, the model assumes linear relationships on the log scale and does not account for potential nonlinearities or interactions, which could be explored in future research.

Despite these limitations, the analysis offers actionable evidence on Toronto's construction cost drivers. More broadly, this research illustrates how municipal open data and regression modeling can inform policy and investment decisions aimed at tackling the city's housing affordability crisis.

Appendix A

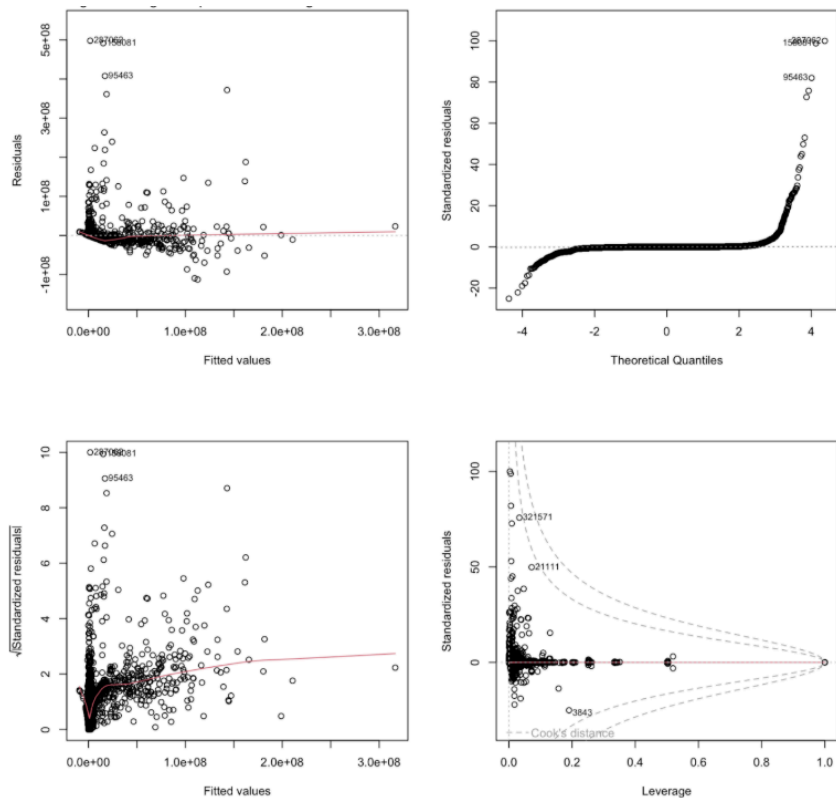


Figure 5. Diagnostic plots for the initial model showing non-normal residuals, heteroscedasticity, and influential points.

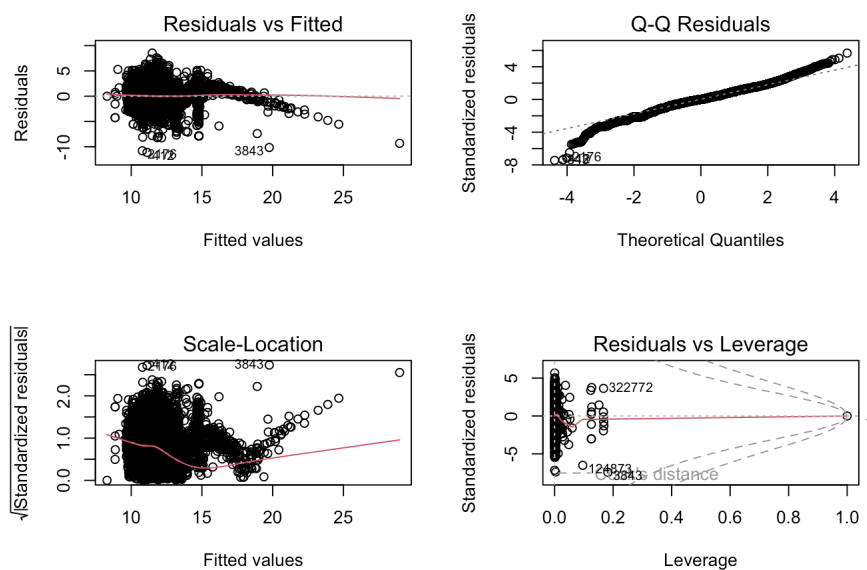


Figure 6. Diagnostic plots for the log-transformed model showing improved residual normality and variance stability.

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